

Module_3: Idiopathic Pulmonary Fibrosis

Team Members:

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Project Title:

Spatial Distribution of Fibrosis in the Murine Bleomycin Model of Lung Fibrotic Disease

Project Goal:

Our company is interested in making a new lung biopsy device.

As a first step, our team has developed an image analysis pipeline that predicts the extent of fibrosis at different biopsy depths.

This project seeks to answer two questions:

1. How does the extent of fibrosis vary throughout the depth of a fibrotic lung?
2. Using interpolation, can we predict the amount of fibrosis at a specific depth into the lung?

Disease Background: Idiopathic Pulmonary Fibrosis

- **Definition** : Idiopathic (cause not known) pulmonary (lung) fibrosis (excess collagen in connective or interstitial tissue).
- **Prevalence & incidence** :
 - Prevalence : 1.25–63 Cases per 100,000 population (More common in North America than Europe) (Rozenberg, et al. 2020)
 - Incidence : 40,000 people diagnosed with IPF Annually in the U.S. (Leonard-Duke, et al., 2025)
- **Risk factors (genetic, lifestyle)** : Increased age, male sex, and smoking history. (Rozenberg, et al. 2020) Genetic can also lead to predisposition such as mutation in Surfactant Protein Related Genes and MUC 5B Promoter and Toll-Interacting Protein and telomere dysfunction. (Duhig 2022)
- **Symptoms** : Dyspnea, Nonproductive Cough, Fatigue, Death (Median Survival of About three years from diagnosis)

- IFP Patient can also develop or have concurrent diseases such as Chronic Obstructive Pulmonary Disease (COPD), Cardiovascular Diseases (Including Pulmonary Hypertension, Venous Thromboembolic Disease, Congestive Heart Failure, Coronary Artery Disease, Gastroesophageal Reflux Disease), Lung Cancer, Obstructive Sleep Apnea, and Mental Health Concerns (Anxiety and Depression). (Though the reason for increase prevalence of comorbidities is not known, perhaps it is all related to aging - telomeres dysfunction.) (Rozenberg, et al. 2020)
- **Standard of care treatment(s) :**
 - Pharmacological Treatment of IPF such as Antifibrotic Drugs : Pirfenidone, Nintedanib, which is shown to attenuate lung function decline. (Still on ongoing field of research, Nintedanib Treatment may also worsen IPF with increased Endothelial Cell Apoptosis and Pericyte-To-Myofibroblast Transition due to Endothelial-Pericyte coupling.)
 - Oxygen Therapy, Pulmonary Rehabilitation, Nutrition
 - Lung Transplant
 - Palliative Care : Medical care focused on providing relief from symptoms, pain, and stress of serious illness to improve quality of life.

(Leonard-Duke, et al., 2025 ; Rozenberg, et al. 2020)
- **Biological mechanisms (anatomy, organ physiology, cell & molecular physiology) :** Following epithelial cell injury, there is an abnormal wound healing process where there is an increased secretion of growth factors (such as Transforming Growth Factor Beta TGFB), increase of fibroblasts and myofibroblasts which lead to interest interstitial fibrosis. (Duhig, 2022)

Data-Set:

Unpublished data was collected by the Peirce-Cottler Lab (Dept. of Biomedical Engineering) and Kim Lab (Division of Pulmonary and Critical Care) at the University of Virginia School of Medicine.

The data is collected from a Bleomycin-Induced Lung Injury Mice Model, where Bleomycin is injected in the the lung of the mcie subject, leading to lung inflammation and massive fibrosis. This is used to model Idiopathic Pulmonary Fibrosis in mice. Afterward, the lung is extracted and then fixed with Paraformaldehyde (so it does not rot) and then mount it in gel or wax to slice with a cryotome. The slides are then mounted on a glass microscope slide, and the immunohistochemistry (IHC) stain with Fluorescent Labeled Antibody which binds to the protein of interest (Utilizing Desmin

Stain to identify Myofibroblasts). They are viewed under a microscope to capture a digital images.

Data Analysis:

1. How does the extent of fibrosis vary throughout the depth of a fibrotic lung?

1.1 Determining Fibrotic Percentage

The original code for this data analysis is written by Prof. Peirce-Cottler, later modified and optimized by Kai-Tang Tseng and Sara Elster to include most data as well as improving efficiency.

The datasets is consisting of black and white pixels with black pixel representing healthy lung tissues and white pixel representing fibrotic tissues.

NOTE : This current datasets can have artifact of vasculature where it incorrectly labels the blood vessels with a bright white ring while it is not a fibrotic tissue. This is the limitation of this dataset.

For our interpolation model, included all data points given as listed in "FileNames and Depths for Students.csv" with the exception of duplicating depths as quadratic and cubic interpolation cannot interpret data with a repeated depth.

```
In [1]: '''Module 3: count black and white pixels and compute the percentage of white pixels'''

from termcolor import colored
import cv2
import numpy as np
import matplotlib.pyplot as plt
from scipy.interpolate import interp1d
import pandas as pd
import os

# Tell python the current directory you are in
directory = os.getcwd()

# Extract data from the FileNames and Depths for Students.csv
data = pd.read_csv(os.path.join(directory, "FileNames and Depths for Students.csv"))

# 1. Define the long column name as a variable to keep code clean
depth_col = 'Depth from lung surface (in micrometers) where image was acquired'

# 2. Filter the ENTIRE DataFrame first
# data_filtered = data[(data[depth_col] < 5000) | (data[depth_col] > 7000)]
data_filtered = data.drop_duplicates(subset=[depth_col], keep='first')

# 3. NOW extract your lists from the filtered version
```

```

depths = data_filtered[depth_col].tolist()
filenames = data_filtered['Filenames'].tolist()

# 4. Build your full paths using the filtered filenames
filenames_complete = [os.path.join(directory, file) for file in filenames]

images = []
white_counts = []
black_counts = []
white_percents = []

# Build the list of all the images you are analyzing

for filename in filenames:
    img = cv2.imread(filename, 0)
    images.append(img)

    _, binary = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)

    white = np.sum(binary == 255)
    black = np.sum(binary == 0)

    white_counts.append(white)
    black_counts.append(black)
    white_percents.append((white/(white+black))*100)

for i in range(len(filenames)):
    print(colored(f'{filenames[i]}:', "red"))
    print()
    print(colored("Pixel count in image:", "yellow"))
    print(colored(f"White pixels in image {filenames[i]}: {white_counts[i]}", "red"))
    print(colored(f"Black pixels in image {filenames[i]}: {black_counts[i]}", "red"))
    print()
    print(colored("Percent white pixels:", "yellow"))
    print(f'{white_percents[i]:.2f}% White | Depth: {depths[i]} microns')
    print()

df = pd.DataFrame({
    "Filenames": filenames,
    "Depths": depths,
    "White percents": white_percents
})

df.to_csv("Percent_White_Pixels.csv", index=False)

print("The .csv file 'Percent_White_Pixels.csv' has been created.")

```

images/MASK_SK658 Llobe ch010039.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010039.jpg: 21648 white

Black pixels in image images/MASK_SK658 Llobe ch010039.jpg: 4172656

Percent white pixels:

0.52% White | Depth: 15 microns

images/MASK_SK658 Llobe ch010021.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010021.jpg: 23900 white

Black pixels in image images/MASK_SK658 Llobe ch010021.jpg: 4170404

Percent white pixels:

0.57% White | Depth: 30 microns

images/MASK_SK658 Llobe ch010017.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010017.jpg: 27561 white

Black pixels in image images/MASK_SK658 Llobe ch010017.jpg: 4166743

Percent white pixels:

0.66% White | Depth: 45 microns

images/MASK_SK658 Llobe ch010036.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010036.jpg: 29677 white

Black pixels in image images/MASK_SK658 Llobe ch010036.jpg: 4164627

Percent white pixels:

0.71% White | Depth: 55 microns

images/MASK_SK658 Llobe ch010019.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010019.jpg: 31331 white

Black pixels in image images/MASK_SK658 Llobe ch010019.jpg: 4162973

Percent white pixels:

0.75% White | Depth: 60 microns

images/MASK_SK658 Llobe ch010022.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010022.jpg: 33151 white

Black pixels in image images/MASK_SK658 Llobe ch010022.jpg: 4161153

Percent white pixels:

0.79% White | Depth: 80 microns

images/MASK_SK658 Llobe ch010018.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010018.jpg: 33746 white

Black pixels in image images/MASK_SK658 Llobe ch010018.jpg: 4160558

Percent white pixels:

0.80% White | Depth: 90 microns

images/MASK_SK658 Llobe ch010023.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010023.jpg: 37508 white

Black pixels in image images/MASK_SK658 Llobe ch010023.jpg: 4156796

Percent white pixels:

0.89% White | Depth: 100 microns

images/MASK_SK658 Llobe ch010030.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010030.jpg: 37799 white

Black pixels in image images/MASK_SK658 Llobe ch010030.jpg: 4156505

Percent white pixels:

0.90% White | Depth: 200 microns

images/MASK_SK658 Slobe ch010156.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010156.jpg: 38068 white

Black pixels in image images/MASK_SK658 Slobe ch010156.jpg: 4156236

Percent white pixels:

0.91% White | Depth: 330 microns

images/MASK_SK658 Llobe ch010040.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010040.jpg: 41943 white

Black pixels in image images/MASK_SK658 Llobe ch010040.jpg: 4152361

Percent white pixels:

1.00% White | Depth: 380 microns

images/MASK_SK658 Llobe ch010051.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010051.jpg: 43424 white

Black pixels in image images/MASK_SK658 Llobe ch010051.jpg: 4150880

Percent white pixels:

1.04% White | Depth: 400 microns

images/MASK_SK658 Llobe ch010032.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010032.jpg: 48667 white

Black pixels in image images/MASK_SK658 Llobe ch010032.jpg: 4145637

Percent white pixels:

1.16% White | Depth: 500 microns

images/MASK_SK658 Slobe ch010048.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010048.jpg: 49491 white

Black pixels in image images/MASK_SK658 Slobe ch010048.jpg: 4144813

Percent white pixels:

1.18% White | Depth: 540 microns

images/MASK_SK658 Llobe ch010025.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010025.jpg: 51040 white

Black pixels in image images/MASK_SK658 Llobe ch010025.jpg: 4143264

Percent white pixels:

1.22% White | Depth: 570 microns

images/MASK_SK658 Llobe ch010061.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010061.jpg: 56003 white

Black pixels in image images/MASK_SK658 Llobe ch010061.jpg: 4138301

Percent white pixels:

1.34% White | Depth: 585 microns

images/MASK_SK658 Llobe ch010024.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010024.jpg: 56360 white

Black pixels in image images/MASK_SK658 Llobe ch010024.jpg: 4137944

Percent white pixels:

1.34% White | Depth: 600 microns

images/MASK_SK658 Slobe ch010157.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010157.jpg: 56419 white

Black pixels in image images/MASK_SK658 Slobe ch010157.jpg: 4137885

Percent white pixels:

1.35% White | Depth: 750 microns

images/MASK_SK658 Llobe ch010035.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010035.jpg: 56534 white

Black pixels in image images/MASK_SK658 Llobe ch010035.jpg: 4137770

Percent white pixels:

1.35% White | Depth: 780 microns

images/MASK_SK658 Llobe ch010171.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010171.jpg: 57531 white

Black pixels in image images/MASK_SK658 Llobe ch010171.jpg: 4136773

Percent white pixels:

1.37% White | Depth: 810 microns

images/MASK_SK658 Slobe ch010159.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010159.jpg: 59426 white

Black pixels in image images/MASK_SK658 Slobe ch010159.jpg: 4134878

Percent white pixels:

1.42% White | Depth: 860 microns

images/MASK_SK658 Slobe ch010158.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010158.jpg: 59788 white

Black pixels in image images/MASK_SK658 Slobe ch010158.jpg: 4134516

Percent white pixels:

1.43% White | Depth: 920 microns

images/MASK_SK658 Llobe ch010031.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010031.jpg: 59908 white

Black pixels in image images/MASK_SK658 Llobe ch010031.jpg: 4134396

Percent white pixels:

1.43% White | Depth: 955 microns

images/MASK_SK658 Slobe ch010066.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010066.jpg: 60715 white

Black pixels in image images/MASK_SK658 Slobe ch010066.jpg: 4133589

Percent white pixels:

1.45% White | Depth: 1000 microns

images/MASK_SK658 Llobe ch010067.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010067.jpg: 62508 white

Black pixels in image images/MASK_SK658 Llobe ch010067.jpg: 4131796

Percent white pixels:

1.49% White | Depth: 1500 microns

images/MASK_SK658 Slobe ch010146.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010146.jpg: 62913 white

Black pixels in image images/MASK_SK658 Slobe ch010146.jpg: 4131391

Percent white pixels:

1.50% White | Depth: 2000 microns

images/MASK_SK658 Llobe ch010164.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010164.jpg: 65021 white

Black pixels in image images/MASK_SK658 Llobe ch010164.jpg: 4129283

Percent white pixels:

1.55% White | Depth: 2200 microns

images/MASK_SK658 Slobe ch010083.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010083.jpg: 65633 white

Black pixels in image images/MASK_SK658 Slobe ch010083.jpg: 4128671

Percent white pixels:

1.56% White | Depth: 2400 microns

images/MASK_SK658 Slobe ch010143.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010143.jpg: 70715 white

Black pixels in image images/MASK_SK658 Slobe ch010143.jpg: 4123589

Percent white pixels:

1.69% White | Depth: 2600 microns

images/MASK_SK658 Llobe ch010026.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010026.jpg: 73224 white

Black pixels in image images/MASK_SK658 Llobe ch010026.jpg: 4121080

Percent white pixels:

1.75% White | Depth: 2800 microns

images/MASK_SK658 Llobe ch010033.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010033.jpg: 76579 white

Black pixels in image images/MASK_SK658 Llobe ch010033.jpg: 4117725

Percent white pixels:

1.83% White | Depth: 2950 microns

images/MASK_SK658 Slobe ch010147.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010147.jpg: 80534 white

Black pixels in image images/MASK_SK658 Slobe ch010147.jpg: 4113770

Percent white pixels:

1.92% White | Depth: 3000 microns

images/MASK_SK658 Llobe ch010174.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010174.jpg: 81014 white

Black pixels in image images/MASK_SK658 Llobe ch010174.jpg: 4113290

Percent white pixels:

1.93% White | Depth: 3100 microns

images/MASK_SK658 Slobe ch010129.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010129.jpg: 82784 white

Black pixels in image images/MASK_SK658 Slobe ch010129.jpg: 4111520

Percent white pixels:

1.97% White | Depth: 3250 microns

images/MASK_SK658 Slobe ch010149.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010149.jpg: 83416 white

Black pixels in image images/MASK_SK658 Slobe ch010149.jpg: 4110888

Percent white pixels:

1.99% White | Depth: 3350 microns

images/MASK_SK658 Llobe ch010065.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010065.jpg: 83951 white

Black pixels in image images/MASK_SK658 Llobe ch010065.jpg: 4110353

Percent white pixels:

2.00% White | Depth: 3900 microns

images/MASK_SK658 Slobe ch010134.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010134.jpg: 91653 white

Black pixels in image images/MASK_SK658 Slobe ch010134.jpg: 4102651

Percent white pixels:

2.19% White | Depth: 4500 microns

images/MASK_SK658 Slobe ch010078.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010078.jpg: 92260 white
Black pixels in image images/MASK_SK658 Slobe ch010078.jpg: 4102044

Percent white pixels:

2.20% White | Depth: 4700 microns

images/MASK_SK658 Slobe ch010110.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010110.jpg: 96012 white
Black pixels in image images/MASK_SK658 Slobe ch010110.jpg: 4098292

Percent white pixels:

2.29% White | Depth: 5300 microns

images/MASK_SK658 Slobe ch010112.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010112.jpg: 97032 white
Black pixels in image images/MASK_SK658 Slobe ch010112.jpg: 4097272

Percent white pixels:

2.31% White | Depth: 5500 microns

images/MASK_SK658 Slobe ch010077.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010077.jpg: 99131 white
Black pixels in image images/MASK_SK658 Slobe ch010077.jpg: 4095173

Percent white pixels:

2.36% White | Depth: 5900 microns

images/MASK_SK658 Llobe ch010168.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010168.jpg: 99350 white
Black pixels in image images/MASK_SK658 Llobe ch010168.jpg: 4094954

Percent white pixels:

2.37% White | Depth: 6000 microns

images/MASK_SK658 Slobe ch010107.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010107.jpg: 100236 white
Black pixels in image images/MASK_SK658 Slobe ch010107.jpg: 4094068

Percent white pixels:

2.39% White | Depth: 6300 microns

images/MASK_SK658 Llobe ch010034.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010034.jpg: 101283 white
Black pixels in image images/MASK_SK658 Llobe ch010034.jpg: 4093021

Percent white pixels:

2.41% White | Depth: 6600 microns

images/MASK_SK658 Slobe ch010126.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010126.jpg: 102335 white

Black pixels in image images/MASK_SK658 Slobe ch010126.jpg: 4091969

Percent white pixels:

2.44% White | Depth: 6800 microns

images/MASK_SK658 Slobe ch010086.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010086.jpg: 103576 white

Black pixels in image images/MASK_SK658 Slobe ch010086.jpg: 4090728

Percent white pixels:

2.47% White | Depth: 6900 microns

images/MASK_SK658 Slobe ch010130.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010130.jpg: 112613 white

Black pixels in image images/MASK_SK658 Slobe ch010130.jpg: 4081691

Percent white pixels:

2.68% White | Depth: 7000 microns

images/MASK_SK658 Slobe ch010142.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010142.jpg: 116432 white

Black pixels in image images/MASK_SK658 Slobe ch010142.jpg: 4077872

Percent white pixels:

2.78% White | Depth: 7100 microns

images/MASK_SK658 Llobe ch010160.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010160.jpg: 118409 white

Black pixels in image images/MASK_SK658 Llobe ch010160.jpg: 4075895

Percent white pixels:

2.82% White | Depth: 7200 microns

images/MASK_SK658 Slobe ch010113.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010113.jpg: 119938 white

Black pixels in image images/MASK_SK658 Slobe ch010113.jpg: 4074366

Percent white pixels:

2.86% White | Depth: 7300 microns

images/MASK_SK658 Slobe ch010063.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010063.jpg: 120910 white

Black pixels in image images/MASK_SK658 Slobe ch010063.jpg: 4073394

Percent white pixels:

2.88% White | Depth: 7400 microns

images/MASK_SK658 Llobe ch010149.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010149.jpg: 124833 white

Black pixels in image images/MASK_SK658 Llobe ch010149.jpg: 4069471

Percent white pixels:

2.98% White | Depth: 7500 microns

images/MASK_SK658 Llobe ch010159.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010159.jpg: 128537 white

Black pixels in image images/MASK_SK658 Llobe ch010159.jpg: 4065767

Percent white pixels:

3.06% White | Depth: 7600 microns

images/MASK_SK658 Slobe ch010059.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010059.jpg: 129607 white

Black pixels in image images/MASK_SK658 Slobe ch010059.jpg: 4064697

Percent white pixels:

3.09% White | Depth: 7700 microns

images/MASK_SK658 Slobe ch010119.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010119.jpg: 137592 white

Black pixels in image images/MASK_SK658 Slobe ch010119.jpg: 4056712

Percent white pixels:

3.28% White | Depth: 8000 microns

images/MASK_SK658 Slobe ch010105.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010105.jpg: 137688 white

Black pixels in image images/MASK_SK658 Slobe ch010105.jpg: 4056616

Percent white pixels:

3.28% White | Depth: 8100 microns

images/MASK_SK658 Slobe ch010088.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010088.jpg: 139549 white

Black pixels in image images/MASK_SK658 Slobe ch010088.jpg: 4054755

Percent white pixels:

3.33% White | Depth: 8200 microns

images/MASK_SK658 Slobe ch010140.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010140.jpg: 139818 white

Black pixels in image images/MASK_SK658 Slobe ch010140.jpg: 4054486

Percent white pixels:

3.33% White | Depth: 8300 microns

images/MASK_SK658 Slobe ch010060.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010060.jpg: 142359 white

Black pixels in image images/MASK_SK658 Slobe ch010060.jpg: 4051945

Percent white pixels:

3.39% White | Depth: 8400 microns

images/MASK_SK658 Llobe ch010146.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010146.jpg: 145829 white

Black pixels in image images/MASK_SK658 Llobe ch010146.jpg: 4048475

Percent white pixels:

3.48% White | Depth: 8500 microns

images/MASK_SK658 Slobe ch010111.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010111.jpg: 150556 white

Black pixels in image images/MASK_SK658 Slobe ch010111.jpg: 4043748

Percent white pixels:

3.59% White | Depth: 8600 microns

images/MASK_SK658 Slobe ch010115.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010115.jpg: 151132 white

Black pixels in image images/MASK_SK658 Slobe ch010115.jpg: 4043172

Percent white pixels:

3.60% White | Depth: 8800 microns

images/MASK_SK658 Llobe ch010121.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Llobe ch010121.jpg: 161111 white

Black pixels in image images/MASK_SK658 Llobe ch010121.jpg: 4033193

Percent white pixels:

3.84% White | Depth: 9000 microns

images/MASK_SK658 Slobe ch010136.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010136.jpg: 163638 white

Black pixels in image images/MASK_SK658 Slobe ch010136.jpg: 4030666

Percent white pixels:

3.90% White | Depth: 9200 microns

images/MASK_SK658 Slobe ch010093.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010093.jpg: 168932 white

Black pixels in image images/MASK_SK658 Slobe ch010093.jpg: 4025372

Percent white pixels:

4.03% White | Depth: 9300 microns

images/MASK_SK658 Slobe ch010096.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010096.jpg: 179894 white

Black pixels in image images/MASK_SK658 Slobe ch010096.jpg: 4014410

Percent white pixels:

4.29% White | Depth: 9400 microns

images/MASK_SK658 Slobe ch010135.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010135.jpg: 186078 white

Black pixels in image images/MASK_SK658 Slobe ch010135.jpg: 4008226

Percent white pixels:

4.44% White | Depth: 9500 microns

images/MASK_SK658 Slobe ch010103.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010103.jpg: 193743 white

Black pixels in image images/MASK_SK658 Slobe ch010103.jpg: 4000561

Percent white pixels:

4.62% White | Depth: 9600 microns

images/MASK_SK658 Slobe ch010104.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010104.jpg: 203857 white

Black pixels in image images/MASK_SK658 Slobe ch010104.jpg: 3990447

Percent white pixels:

4.86% White | Depth: 9700 microns

images/MASK_SK658 Slobe ch010068.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010068.jpg: 214773 white

Black pixels in image images/MASK_SK658 Slobe ch010068.jpg: 3979531

Percent white pixels:

5.12% White | Depth: 9800 microns

images/MASK_SK658 Slobe ch010097.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010097.jpg: 221302 white

Black pixels in image images/MASK_SK658 Slobe ch010097.jpg: 3973002

Percent white pixels:

5.28% White | Depth: 9900 microns

images/MASK_SK658 Slobe ch010092.jpg:

Pixel count in image:

White pixels in image images/MASK_SK658 Slobe ch010092.jpg: 328891 white

Black pixels in image images/MASK_SK658 Slobe ch010092.jpg: 3865413

Percent white pixels:

7.84% White | Depth: 10000 microns

The .csv file 'Percent_White_Pixels.csv' has been created.

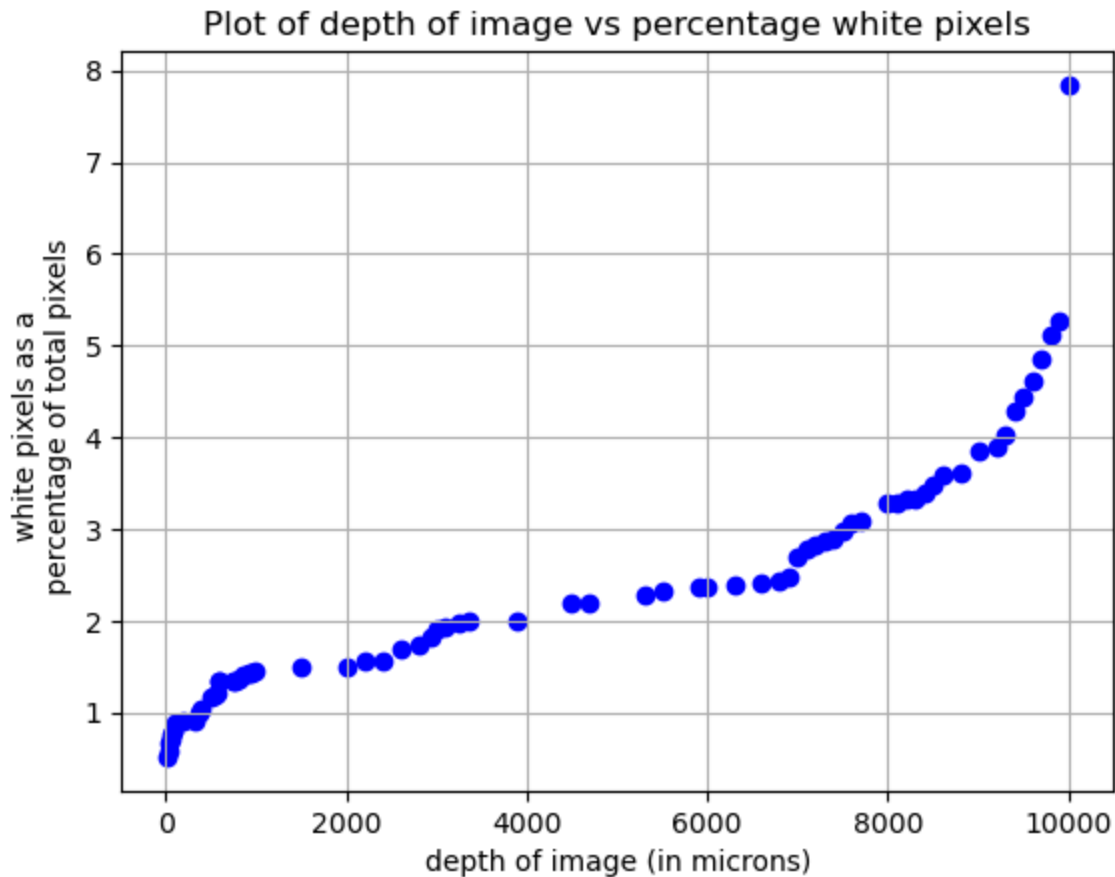
1.2 Graphing our data

```
In [2]: x = depths
        y = white_percents

        depths_i = depths[:]
        white_percents_i = white_percents[:]

        plt.scatter(depths, white_percents, marker='o', linestyle='--', color='blue')
        plt.title('Plot of depth of image vs percentage white pixels')
        plt.xlabel('depth of image (in microns)')
        plt.ylabel('white pixels as a \n percentage of total pixels')
        plt.grid(True)

        # Adjust layout to prevent overlap
        plt.show()
```

2. Using interpolation, can we predict the amount of fibrosis at a specific depth into the lung?

The original code for this data analysis is written by Prof. Peirce-Cottler.

Interpolation is the process of approximating the location of a data point based on other given data.

2.1 Interpolating Linear

The linear Interpolating Point chosen for the follow graph was a depth of 2345, obtained using calculation from two points.

```
In [3]: # Interpolate a point: given a depth, find the corresponding white pixel per
interpolate_depth = float(input(colored(
    "Enter the depth at which you want to interpolate a point (in microns):
x = depths
y = white_percents

# You can also use 'quadratic', 'cubic', etc.
i = interp1d(x, y, kind='linear')
interpolate_point = i(interpolate_depth)
print(colored(
    f'The interpolated point is at the x-coordinate {interpolate_depth} and y-co
```

```

depths_i = depths[:]
depths_i.append(interpolate_depth)
white_percents_i = white_percents[:]
white_percents_i.append(interpolate_point)

# make two plots: one that doesn't contain the interpolated point, just the
fig, axs = plt.subplots(2, 1)

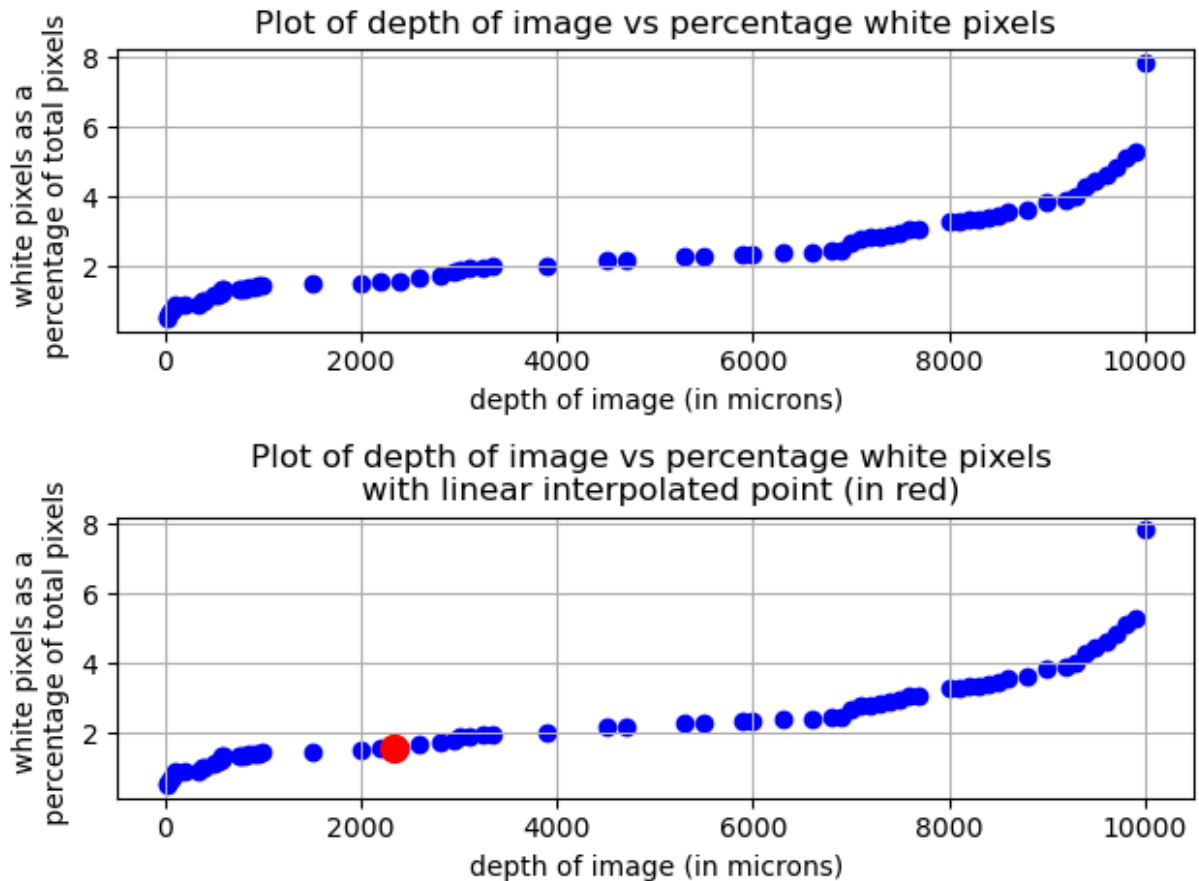
axs[0].scatter(depths, white_percents, marker='o', linestyle='-', color='blue')
axs[0].set_title('Plot of depth of image vs percentage white pixels')
axs[0].set_xlabel('depth of image (in microns)')
axs[0].set_ylabel('white pixels as a \n percentage of total pixels')
axs[0].grid(True)

axs[1].scatter(depths_i, white_percents_i, marker='o',
               linestyle='-', color='blue')
axs[1].set_title(
    'Plot of depth of image vs percentage white pixels \n with linear inter
axs[1].set_xlabel('depth of image (in microns)')
axs[1].set_ylabel('white pixels as a \n percentage of total pixels')
axs[1].grid(True)
axs[1].scatter(depths_i[len(depths_i)-1], white_percents_i[len(white_percent
                    color='red', s=100, label='Highlighted point')

# Adjust layout to prevent overlap
plt.tight_layout()
plt.show()

```

The interpolated point is at the x-coordinate 2345.0 and y-coordinate 1.5608000755310059.



2.2 Interpolating Quadratic

The quadratic Interpolating Point chosen for the follow graph was a depth of 2345, obtained using calculation from three points.

```
In [4]: # Interpolate a point: given a depth, find the corresponding white pixel per
interpolate_depth = float(input(colored(
    "Enter the depth at which you want to interpolate a point (in microns):
x = depths
y = white_percents

# You can also use 'quadratic', 'cubic', etc.
i = interp1d(x, y, kind='quadratic')
interpolate_point = i(interpolate_depth)
print(colored(
    f'The interpolated point is at the x-coordinate {interpolate_depth} and y-co

depths_i = depths[:]
depths_i.append(interpolate_depth)
white_percents_i = white_percents[:]
white_percents_i.append(interpolate_point)

# make two plots: one that doesn't contain the interpolated point, just the
fig, axs = plt.subplots(2, 1)
```

```

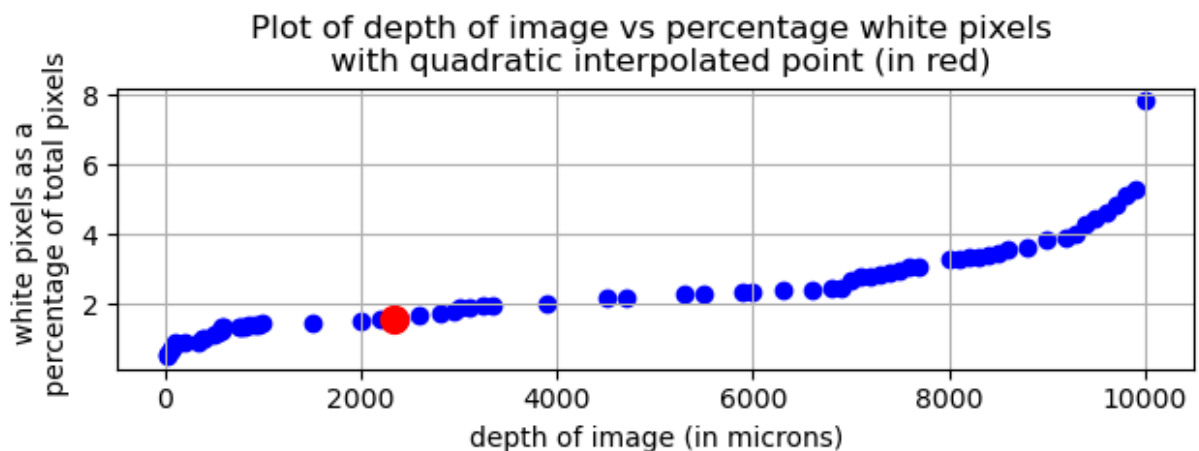
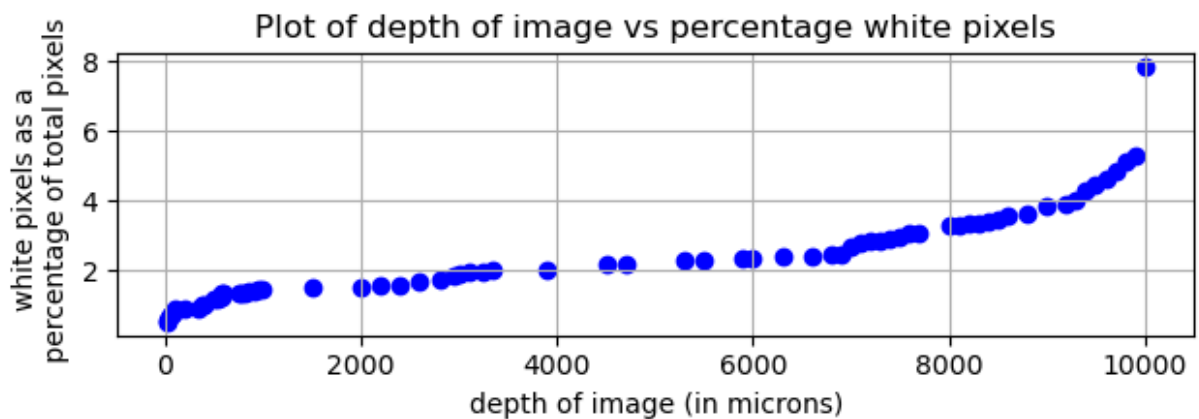
axs[0].scatter(depths, white_percents, marker='o', linestyle='-', color='blue')
axs[0].set_title('Plot of depth of image vs percentage white pixels')
axs[0].set_xlabel('depth of image (in microns)')
axs[0].set_ylabel('white pixels as a \n percentage of total pixels')
axs[0].grid(True)

axs[1].scatter(depths_i, white_percents_i, marker='o',
               linestyle='-', color='blue')
axs[1].set_title('Plot of depth of image vs percentage white pixels \n with quadratic interpolation')
axs[1].set_xlabel('depth of image (in microns)')
axs[1].set_ylabel('white pixels as a \n percentage of total pixels')
axs[1].grid(True)
axs[1].scatter(depths_i[len(depths_i)-1], white_percents_i[len(white_percents_i)-1],
               color='red', s=100, label='Highlighted point')

# Adjust layout to prevent overlap
plt.tight_layout()
plt.show()

```

The interpolated point is at the x-coordinate 2345.0 and y-coordinate 1.5523349828950628.



2.3 Interpolating Cubic

The Cubic Interpolating Point chosen for the follow graph was a depth of 2345, obtained using calculation from three points.

```
In [5]: # Interpolate a point: given a depth, find the corresponding white pixel per

interpolate_depth = float(input(colored(
    "Enter the depth at which you want to interpolate a point (in microns):

x = depths
y = white_percents

# You can also use 'quadratic', 'cubic', etc.
i = interp1d(x, y, kind='cubic')
interpolate_point = i(interpolate_depth)
print(colored(
    f'The interpolated point is at the x-coordinate {interpolate_depth} and y-co

depths_i = depths[:]
depths_i.append(interpolate_depth)
white_percents_i = white_percents[:]
white_percents_i.append(interpolate_point)

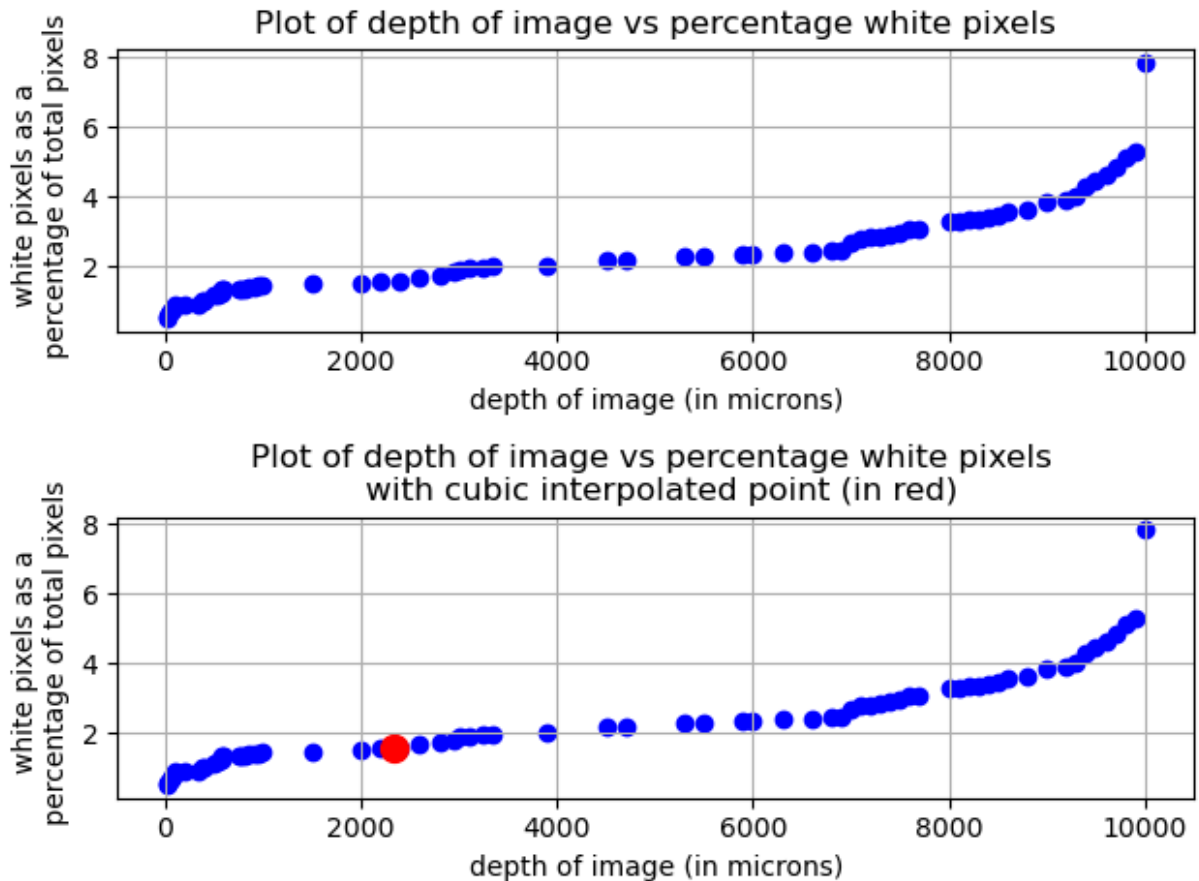
# make two plots: one that doesn't contain the interpolated point, just the
fig, axs = plt.subplots(2, 1)

axs[0].scatter(depths, white_percents, marker='o', linestyle='-', color='blue')
axs[0].set_title('Plot of depth of image vs percentage white pixels')
axs[0].set_xlabel('depth of image (in microns)')
axs[0].set_ylabel('white pixels as a \n percentage of total pixels')
axs[0].grid(True)

axs[1].scatter(depths_i, white_percents_i, marker='o',
               linestyle='-', color='blue')
axs[1].set_title(
    'Plot of depth of image vs percentage white pixels \n with cubic interp
axs[1].set_xlabel('depth of image (in microns)')
axs[1].set_ylabel('white pixels as a \n percentage of total pixels')
axs[1].grid(True)
axs[1].scatter(depths_i[len(depths_i)-1], white_percents_i[len(white_percent
               color='red', s=100, label='Highlighted point')

# Adjust layout to prevent overlap
plt.tight_layout()
plt.show()
```

The interpolated point is at the x-coordinate 2345.0 and y-coordinate 1.5533846763089287.



2.4 Validation of Interpolated Points

We perform the interpolations with three methods : linear, quadratic, and cubic interpolations. We excluded dataset from a depth of 5000-7000 to test our interpolation results.

Here we perform a calculation of interpolation at depths of 5500, 6000, and 6600 using the different methods in comparison with the actual result.

```
In [6]: # Interpolate at three test points for three different method, compare to actual
# Filter depths and white percents lists to include depths excluding 5000-7000
new_depths = []
new_white_percents = []

for i in range(len(depths)):
    if (depths[i] < 5000) or (depths[i] > 7000):
        new_depths.append(depths[i])
        new_white_percents.append(white_percents[i])

x = new_depths
y = new_white_percents

# Test point 1: depth = 5500; actual percent = 2.3134231567382812
interpolate_depth = float(5500)
actual_percent = 2.3134231567382812
```

```

# Linear
i = interp1d(x, y, kind='linear')
interpolate_point_linear = i(interpolate_depth)
linear_abs_error = abs(actual_percent - interpolate_point_linear)
linear_rel_error = (linear_abs_error/actual_percent) * 100
print(colored(f'Test Point 1 Depth: {interpolate_depth}', 'yellow'))
print(colored(f'Actual % white pixels: {actual_percent}%', 'blue'))
print(colored(f'% white pixels (linear interpolation): {interpolate_point_linear}', 'blue'))
print(colored(f'Linear Interpolation Absolute Error: {linear_abs_error}.', 'red'))
print(colored(f'Linear Interpolation Relative Error: {linear_rel_error}%', 'red'))

# Quadratic
i = interp1d(x, y, kind='quadratic')
interpolate_point_quadratic = i(interpolate_depth)
quad_abs_error = abs(actual_percent - interpolate_point_quadratic)
quad_rel_error = (quad_abs_error/actual_percent) * 100
print(colored(f'% white pixels (quadratic interpolation): {interpolate_point_quadratic}', 'blue'))
print(colored(f'Quadratic Interpolation Absolute Error: {quad_abs_error}.', 'red'))
print(colored(f'Quadratic Interpolation Relative Error: {quad_rel_error}%', 'red'))

# Cubic
i = interp1d(x, y, kind='cubic')
interpolate_point_cubic = i(interpolate_depth)
cub_abs_error = abs(actual_percent - interpolate_point_cubic)
cub_rel_error = (cub_abs_error/actual_percent) * 100
print(colored(f'% white pixels (cubic interpolation): {interpolate_point_cubic}', 'blue'))
print(colored(f'Quadratic Interpolation Absolute Error: {cub_abs_error}.', 'red'))
print(colored(f'Quadratic Interpolation Relative Error: {cub_rel_error}%', 'red'))

# Test point 2: depth = 6000; actual percent = 2.3686885833740234
interpolate_depth = float(6000)
actual_percent = 2.3686885833740234

# Linear
i = interp1d(x, y, kind='linear')
interpolate_point_linear = i(interpolate_depth)
linear_abs_error = abs(actual_percent - interpolate_point_linear)
linear_rel_error = (linear_abs_error/actual_percent) * 100
print(colored(f'Test Point 2 Depth: {interpolate_depth}', 'yellow'))
print(colored(f'Actual % white pixels: {actual_percent}%', 'blue'))
print(colored(f'% white pixels (linear interpolation): {interpolate_point_linear}', 'blue'))
print(colored(f'Linear Interpolation Absolute Error: {linear_abs_error}.', 'red'))
print(colored(f'Linear Interpolation Relative Error: {linear_rel_error}%', 'red'))

# Quadratic
i = interp1d(x, y, kind='quadratic')
interpolate_point_quadratic = i(interpolate_depth)
quad_abs_error = abs(actual_percent - interpolate_point_quadratic)
quad_rel_error = (quad_abs_error/actual_percent) * 100
print(colored(f'% white pixels (quadratic interpolation): {interpolate_point_quadratic}', 'blue'))
print(colored(f'Quadratic Interpolation Absolute Error: {quad_abs_error}.', 'red'))
print(colored(f'Quadratic Interpolation Relative Error: {quad_rel_error}%', 'red'))

```

```

# Cubic
i = interp1d(x, y, kind='cubic')
interpolate_point_cubic = i(interpolate_depth)
cub_abs_error = abs(actual_percent - interpolate_point_quadratic)
cub_rel_error = (cub_abs_error/actual_percent) * 100
print(colored(f'% white pixels (cubic interpolation): {interpolate_point_cubic}', 'white'))
print(colored(f'Quadratic Interpolation Absolute Error: {cub_abs_error}.', 'red'))
print(colored(f'Quadratic Interpolation Relative Error: {cub_rel_error}%.\n', 'red'))

# Test point 3: depth = 6600; actual percent = 2.4147748947143555
interpolate_depth = float(6600)
actual_percent = 2.4147748947143555

# Linear
i = interp1d(x, y, kind='linear')
interpolate_point_linear = i(interpolate_depth)
linear_abs_error = abs(actual_percent - interpolate_point_linear)
linear_rel_error = (linear_abs_error/actual_percent) * 100
print(colored(f'Test Point 3 Depth: {interpolate_depth}', 'yellow'))
print(colored(f'Actual % white pixels: {actual_percent}%', 'blue'))
print(colored(f'% white pixels (linear interpolation): {interpolate_point_linear}', 'white'))
print(colored(f'Linear Interpolation Absolute Error: {linear_abs_error}.', 'red'))
print(colored(f'Linear Interpolation Relative Error: {linear_rel_error}%.\n', 'red'))

# Quadratic
i = interp1d(x, y, kind='quadratic')
interpolate_point_quadratic = i(interpolate_depth)
quad_abs_error = abs(actual_percent - interpolate_point_quadratic)
quad_rel_error = (quad_abs_error/actual_percent) * 100
print(colored(f'% white pixels (quadratic interpolation): {interpolate_point_quadratic}', 'white'))
print(colored(f'Quadratic Interpolation Absolute Error: {quad_abs_error}.', 'red'))
print(colored(f'Quadratic Interpolation Relative Error: {quad_rel_error}%.\n', 'red'))

# Cubic
i = interp1d(x, y, kind='cubic')
interpolate_point_cubic = i(interpolate_depth)
cub_abs_error = abs(actual_percent - interpolate_point_quadratic)
cub_rel_error = (cub_abs_error/actual_percent) * 100
print(colored(f'% white pixels (cubic interpolation): {interpolate_point_cubic}', 'white'))
print(colored(f'Quadratic Interpolation Absolute Error: {cub_abs_error}.', 'red'))
print(colored(f'Quadratic Interpolation Relative Error: {cub_rel_error}%.\n', 'red'))

```


Test Point 1 Depth: 5500.0
 Actual % white pixels: 2.3134231567382812%.
 % white pixels (linear interpolation): 2.3917516072591147.
 Linear Interpolation Absolute Error: 0.07832845052083348.
 Linear Interpolation Relative Error: 3.3858246076895657%.
 % white pixels (quadratic interpolation): 2.2870875502757326.
 Quadratic Interpolation Absolute Error: 0.026335606462548622.
 Quadratic Interpolation Relative Error: 1.1383825905710852%.
 % white pixels (cubic interpolation): 2.278217586036909.
 Quadratic Interpolation Absolute Error: 0.026335606462548622.
 Quadratic Interpolation Relative Error: 1.1383825905710852%.

Test Point 2 Depth: 6000.0
 Actual % white pixels: 2.3686885833740234%.
 % white pixels (linear interpolation): 2.5118152300516763.
 Linear Interpolation Absolute Error: 0.14312664667765285.
 Linear Interpolation Relative Error: 6.042442543197444%.
 % white pixels (quadratic interpolation): 2.384954369248895.
 Quadratic Interpolation Absolute Error: 0.016265785874871597.
 Quadratic Interpolation Relative Error: 0.6867000579579007%.
 % white pixels (cubic interpolation): 2.381554553412111.
 Quadratic Interpolation Absolute Error: 0.016265785874871597.
 Quadratic Interpolation Relative Error: 0.6867000579579007%.

Test Point 3 Depth: 6600.0
 Actual % white pixels: 2.4147748947143555%.
 % white pixels (linear interpolation): 2.6558915774027505.
 Linear Interpolation Absolute Error: 0.24111668268839503.
 Linear Interpolation Relative Error: 9.985058367807689%.
 % white pixels (quadratic interpolation): 2.5657304798788996.
 Quadratic Interpolation Absolute Error: 0.1509555851645441.
 Quadratic Interpolation Relative Error: 6.251331562828787%.
 % white pixels (cubic interpolation): 2.566626826119028.
 Quadratic Interpolation Absolute Error: 0.1509555851645441.
 Quadratic Interpolation Relative Error: 6.251331562828787%.

3. Analysis of Blood Vessel Proximal and Distal Density

When looking at the images, we observed that around bright white ring (which may signify an artifact of the blood vessels), we realized there seemsto be another black ring that encase this white ring. In another word, we think that perhaps the white density % is lower closer (proximal) to the blood vessel artifact than further away (distal). Rather than just saying we observe this phenomena, we want to be able to quantify it and analysis it's satistical significance.

3.1 Determine the Proximal and Distal Fibrosis Density

The code generated to determine the Proximal and Distal Fibrosis Density is largely written by Google's AI Model: Gemini 3.

The logic is to look through the image for circular shape and have a hole. This marks it as having vessel and label it with a red circle on an outputed image. It then create a

boundary around it of 4500 pixels in all directions, this encases the proximal range (denoted by the orange circle). The Proximal Density is calculated as the density inside the orange ring and outside the red ring. With the Distal Density is just everything outside the orange ring.

Here the dialted boundaries is set to 4500 pixels, this is hard coded and we realized there is significant limitations of this. One if the black ring around it is smaller than 4500 pixels, the proximal density is elavated, if the black ring around it is bigger than the distal fibrosis is depressed. This is a major limitation of this analysis. But for the majority of the image selected, it captures the majority of the "black ring" region for analysis.

```
In [7]: # --- SETUP ---
Location = os.getcwd()

filenames_raw = [
    "MASK_Sk658 Llobe ch010026.jpg",
    "MASK_Sk658 Llobe ch010067.jpg",
    "MASK_Sk658 Llobe ch010034.jpg",
    "MASK_SK658 Slobe ch010087.jpg",
    "MASK_SK658 Slobe ch010118.jpg",
    "MASK_Sk658 Llobe ch010146.jpg"
]

# Create full paths
filenames = [os.path.join(Location, "images", f) for f in filenames_raw]
results_list = []
proximal_list = []
distal_list = []

# --- PROCESSING LOOP ---
for full_path in filenames:
    # Load image as grayscale
    img = cv2.imread(full_path, 0)

    if img is None:
        print(colored(f"Could not load: {full_path}", "red"))
        continue

    # 1. CLEANING THE IMAGE
    # Median blur removes small white noise/specks
    denoised = cv2.medianBlur(img, 5)
    _, binary = cv2.threshold(denoised, 127, 255, cv2.THRESH_BINARY)

    # Morphological Closing bridges small gaps in the vessel walls
    kernel_bridge = np.ones((15, 15), np.uint8)
    closed = cv2.morphologyEx(binary, cv2.MORPH_CLOSE, kernel_bridge)

    # 2. FINDING THE VESSELS (RED CIRCLES)
    contours, hierarchy = cv2.findContours(closed, cv2.RETR_CCOMP, cv2.CHAINLIST)

    vessel_mask = np.zeros_like(img)
    # Create a color version for plotting (BGR to RGB for Matplotlib)
```

```

display_img = cv2.cvtColor(img, cv2.COLOR_GRAY2RGB)
vessels_found = 0

if hierarchy is not None:
    for idx, cnt in enumerate(contours):
        area = cv2.contourArea(cnt)
        perimeter = cv2.arcLength(cnt, True)
        if perimeter == 0 or area < 1500: continue

        # Logic to identify a vessel (roundness and presence of a hole/
        circularity = 4 * np.pi * (area / (perimeter * perimeter))
        has_hole = hierarchy[0][idx][2] != -1

        if (has_hole and circularity > 0.2) or (area > 5000):
            vessels_found += 1
            # Fill the vessel in our internal mask
            cv2.drawContours(vessel_mask, [cnt], -1, 255, -1)
            # Draw the RED identification circle on the display image
            cv2.drawContours(display_img, [cnt], -1, (255, 0, 0), 5) # F

# 3. CREATING THE PROXIMITY REGION (ORANGE RING)
if vessels_found > 0:
    # Dilate (expand) the vessel mask by 0 pixels to create the outer bo
    kernel_dil = np.ones((300, 300), np.uint8)
    dilated = cv2.dilate(vessel_mask, kernel_dil, iterations=1)

    # Subtract the vessel from the dilated area to get only the ring arc
    proximal_mask = cv2.subtract(dilated, vessel_mask)

    # Draw the ORANGE proximity circle on the display image
    ring_contours, _ = cv2.findContours(proximal_mask, cv2.RETR_EXTERNAL
    cv2.drawContours(display_img, ring_contours, -1, (255, 165, 0), 3) #

# 4. DENSITY CALCULATIONS
_, raw_binary = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)

# Count white pixels (fibrosis) inside the orange ring
proximal_white = np.sum((raw_binary == 255) & (proximal_mask == 255))
proximal_total = np.sum(proximal_mask == 255)
proximal_dens = (proximal_white / proximal_total * 100) if proximal_

# Count white pixels in the Parenchyma (everywhere else)
excluded_space = cv2.bitwise_or(vessel_mask, proximal_mask)
distal_mask = cv2.bitwise_not(excluded_space)
distal_white = np.sum((raw_binary == 255) & (distal_mask == 255))
distal_total = np.sum(distal_mask == 255)
distal_dens = (distal_white / distal_total * 100) if distal_total >
else:
    proximal_dens, distal_dens = -1, -1

# Store results
results_list.append({
    "Filename": os.path.basename(full_path),
    "Proximal_Density_%": round(proximal_dens, 2),
    "Distal_Density_%": round(distal_dens, 2)
})

```

```

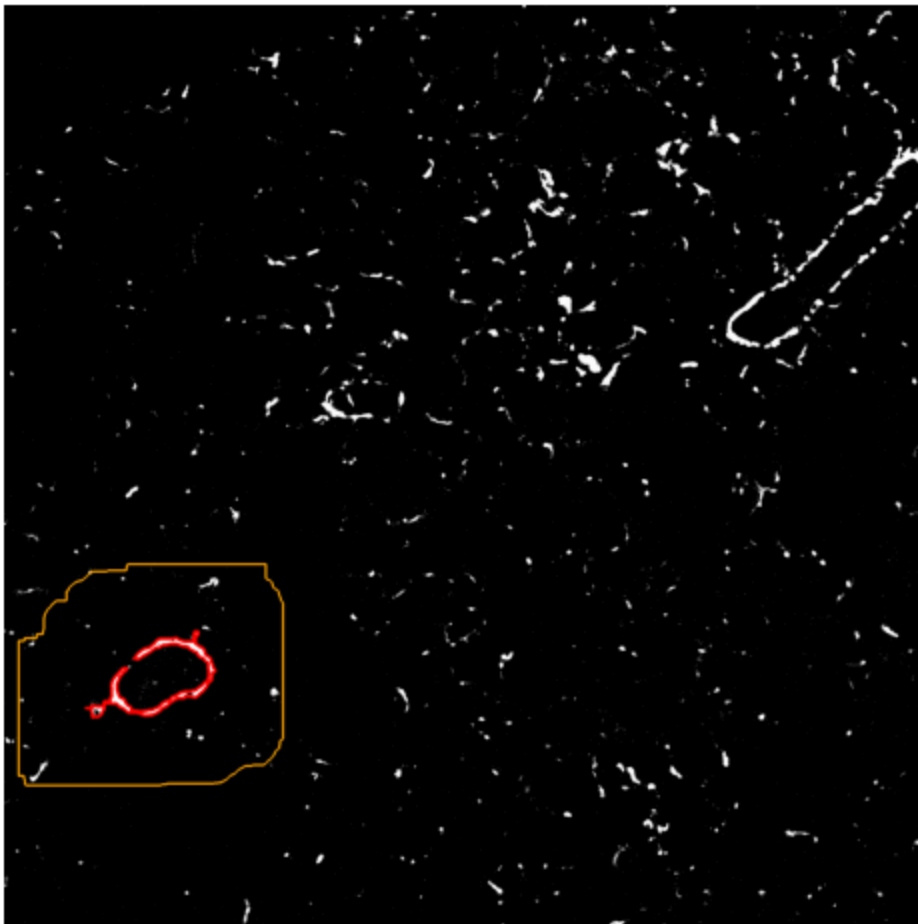
proximal_list.append(proximal_dens)
distal_list.append(distal_dens)

# --- PLOT THE RESULT ---
plt.figure(figsize=(10, 6))
plt.imshow(display_img)
plt.title(f'Analysis: {os.path.basename(full_path)}\nRed = Vessel | Orange = Perivascular Zone (60px)')
plt.axis('off')
plt.show()

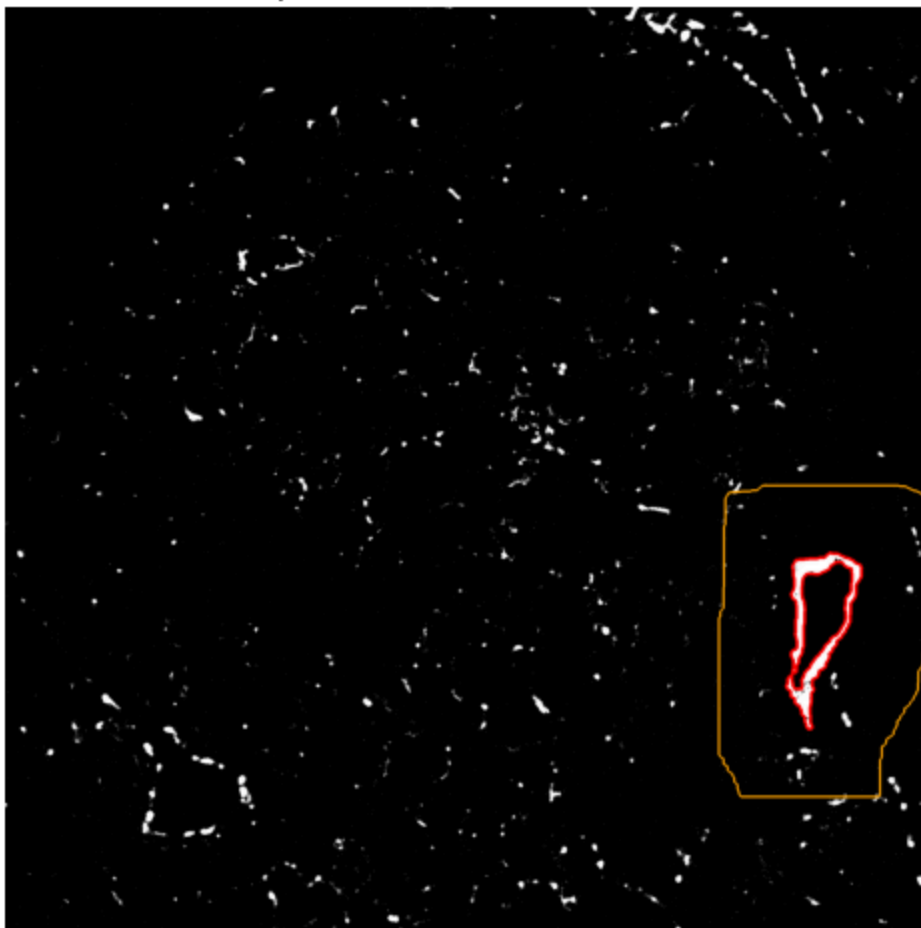
# --- DISPLAY DATA TABLE ---
df_results = pd.DataFrame(results_list)
display(df_results)

```

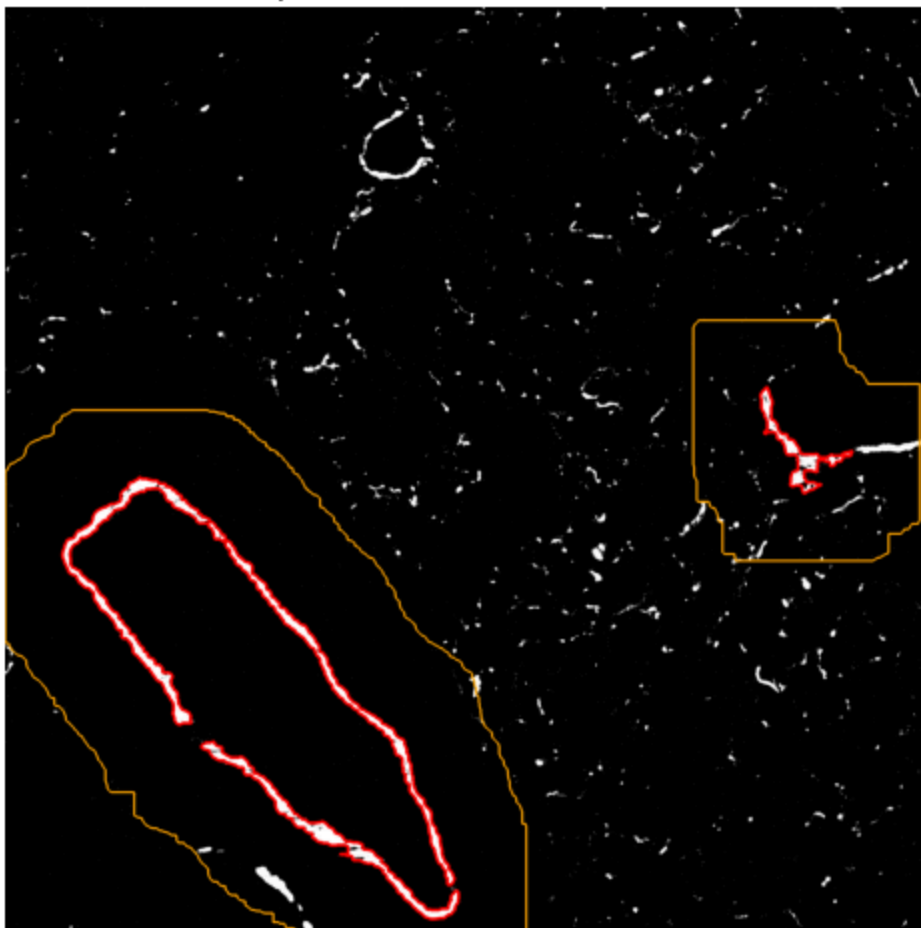
Analysis: MASK_Sk658 Llobe ch010026.jpg
 Red = Vessel | Orange = Perivascular Zone (60px)



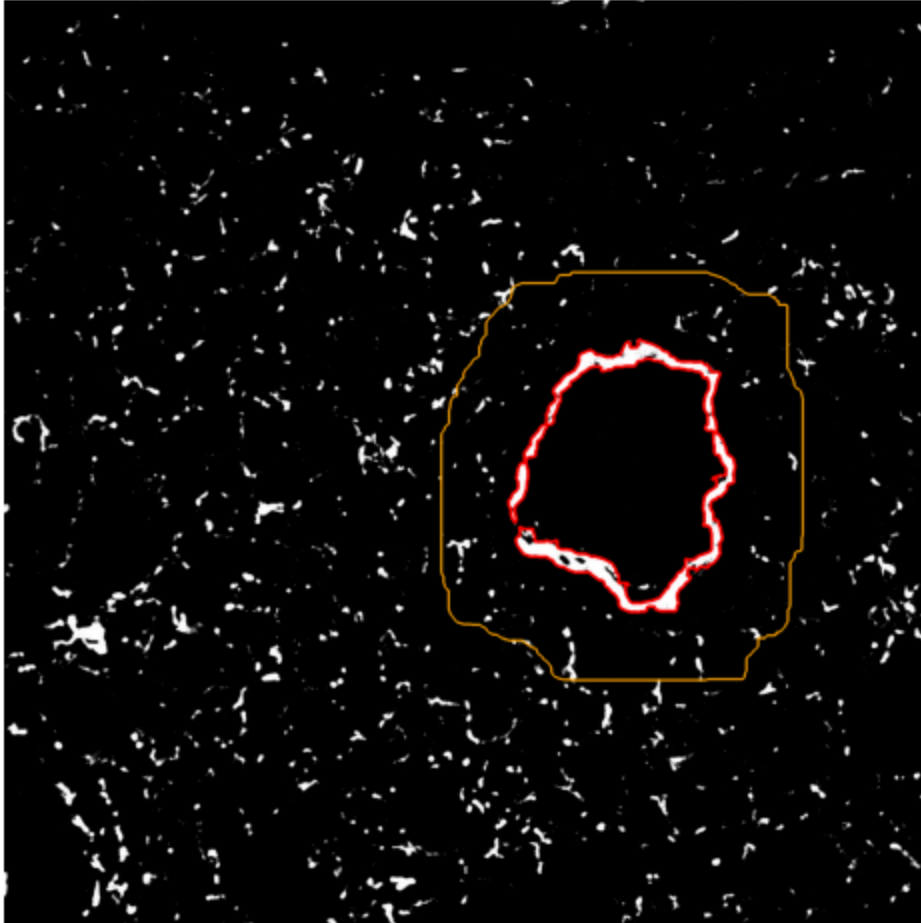
Analysis: MASK_Sk658 Llobe ch010067.jpg
Red = Vessel | Orange = Perivascular Zone (60px)



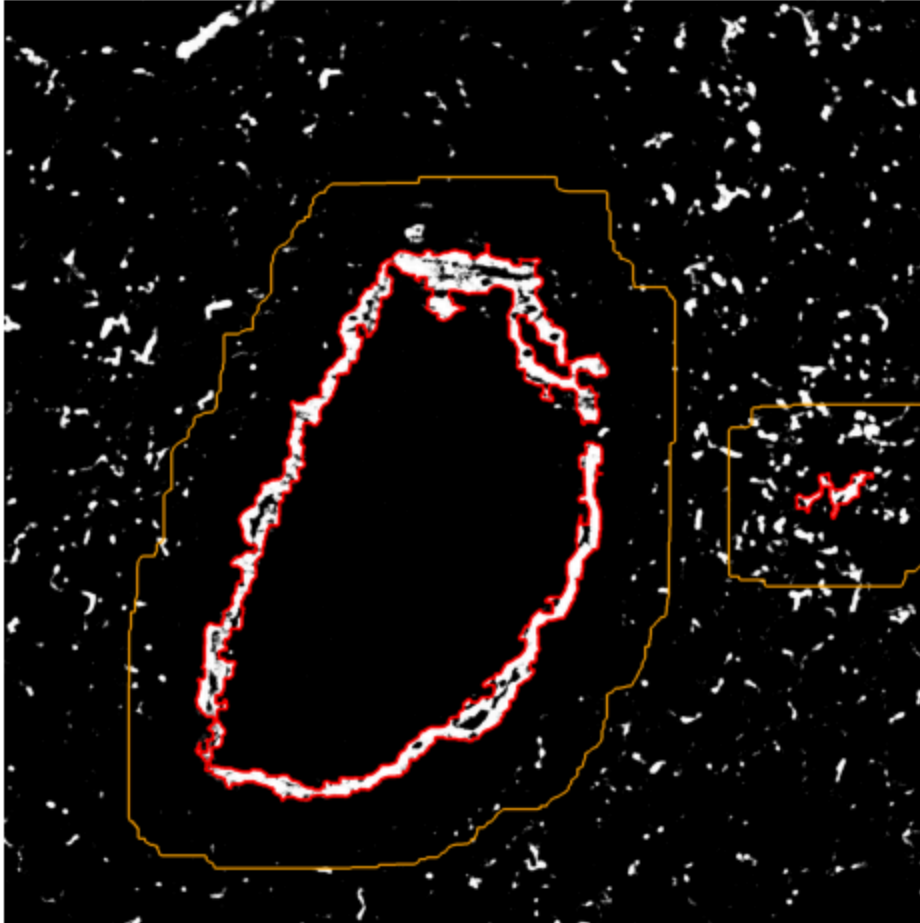
Analysis: MASK_Sk658 Llobe ch010034.jpg
Red = Vessel | Orange = Perivascular Zone (60px)



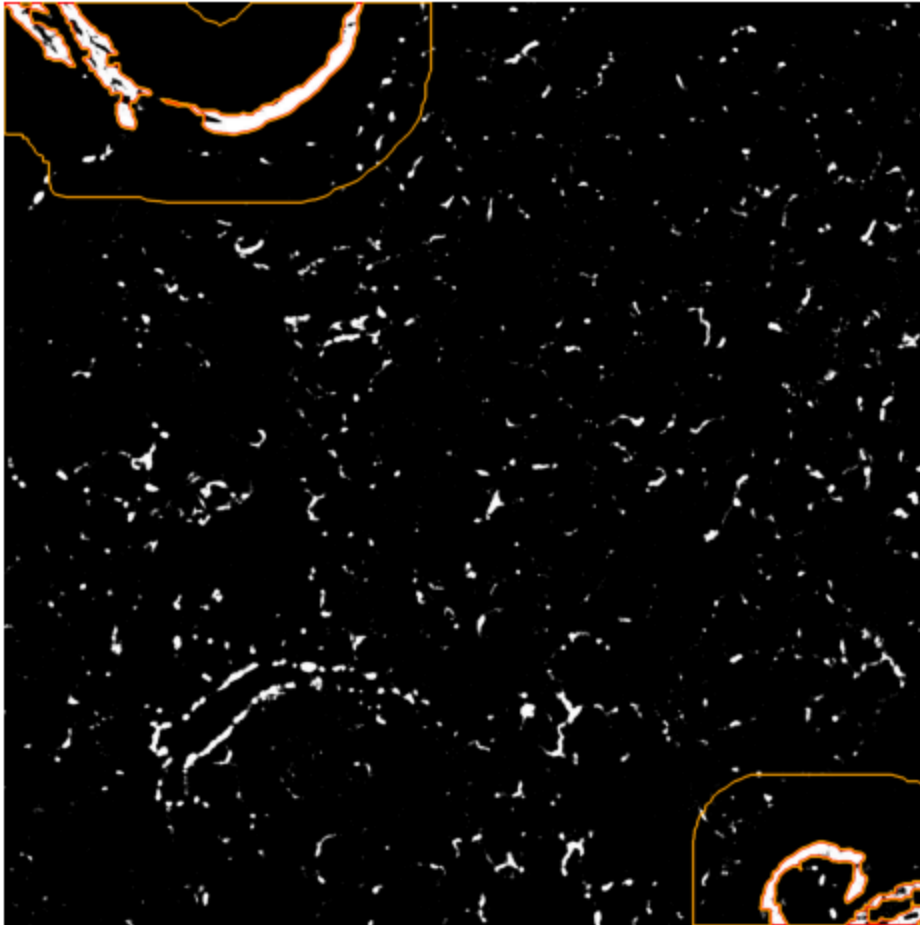
Analysis: MASK_SK658 Slobe ch010087.jpg
Red = Vessel | Orange = Perivascular Zone (60px)



Analysis: MASK_SK658 Slobe ch010118.jpg
Red = Vessel | Orange = Perivascular Zone (60px)



Analysis: MASK_Sk658 Llobe ch010146.jpg
 Red = Vessel | Orange = Perivascular Zone (60px)



	Filename	Proximal_Density_%	Distal_Density_%
0	MASK_Sk658 Llobe ch010026.jpg	0.99	1.63
1	MASK_Sk658 Llobe ch010067.jpg	1.08	1.14
2	MASK_Sk658 Llobe ch010034.jpg	0.97	1.25
3	MASK_SK658 Slobe ch010087.jpg	1.64	3.00
4	MASK_SK658 Slobe ch010118.jpg	1.97	3.42
5	MASK_Sk658 Llobe ch010146.jpg	1.37	2.45

3.2 Statistical Significance

Since the Proximal and Distal Density data is related (they both came from the same image in the same density), we implement a Paired T-Test.

```
In [8]: from scipy import stats

# Use the PAIRED T-Test because both measurements come from the same image
t_stat, p_val = stats.ttest_rel(proximal_list, distal_list)

print(f"Paired T-Test P-Value: {p_val:.4f}")
```

```
#Check for Significance
if p_val < 0.05:
    print(colored("RESULT: SIGNIFICANT DIFFERENCE", "green", attrs=['bold']))
    print("The fibrosis levels are statistically different between the two r
else:
    print(colored("RESULT: NOT SIGNIFICANT", "red", attrs=['bold']))
```

Paired T-Test P-Value: 0.0181

RESULT: SIGNIFICANT DIFFERENCE

The fibrosis levels are statistically different between the two regions.

Verify and validate your analysis:

To validate our interpolation models, we excluded a portion of the actual data during testing (depth of 5000-7000) to test interpolation for a range within. We obtain actual data (calculation for white pixel percentage of the actual image) and interpolation data and compared them.

We realized that Quadratic and Cubic Interpolation generate similar errors, outperforming Linear Interpolation. Quadratic and Cubic Interpolation have a relative error of 0.6% to about 6% which is a relatively small percentage validating interpolation data given a certain range of error.

Conclusions :

Based on this project, we successfully show that in mice lung fibrosis model, as depth is increased, percentage of fibrosis also increased. We also created an interpolation program that is shown to accurately predict percentage of fibrosis given the depth with a range of error.

Ethical Implications :

Mice model were used in this research, we are analyzing already collected data of mice model, optimizing the three "R" of animal research. In this particular research, "Reduction" is optimized by extracting even more data from one mice model allowing this model to generate more information from the same number of animal used (by analyzing already collected data rather than new data).

It is also important to note that the model is based on mice lung fibrosis which have a bit different lung physiology and anatomy compared to human thus more research is needed before making a generalization of depths and fibrosis percentage of human idiopathic lung fibrosis. Furthermore, while the interpolation has a relatively low error, it is important to understand that it is still an approximation and still has a source of error.

Limitations and Future Work:

Limitations of this work is that we only have fibrosis data from this mice, given more data with IHC staining we can analysis even more comprehensively into the interaction of lung fibrosis in this mice model. Second our verification of interpolation can also be optimized with less excluded data point and excluding only the specific point for testing to ensure a more accurate interpolation.

Regarding observation of less fibrotic precentage around blood vessel, the code for detecting can be optimized to detects vessel better (it is missing some currenty and misidentifying non vessel as vessel). Furthermore the range for proximal can also be optimize as right now it is hard coded to be certain distance while some vessel have smaller and some vessel have large black ring around it. The causes the proximal and distal precentage to be more unprecise.

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